
SM131 – Time Value of Money Practice Problems

v 1.0, December 8, 2025; Jeff Furman (all errors are his responsibility)

Note #1: Questions 1-4 are relatively straightforward questions involving single cash flows; Questions 5-8 involve single outflows and single inflows. Questions 9-12 involve multiple outflows and multiple inflows.

Note #2: Practice questions appear without answers on pages 1-2; Practice questions with answers appear on pages 3-6.

Q1-Q4: Single cash flow questions

PV of a single future payoff

- (1)** Suppose you expect to receive a \$500 bonus from your internship three years from today. If the appropriate discount rate is 10% per year, what is the present value today of that \$500?

FV of a single investment

- (2)** On January 1, 2025, you deposit \$800 into a savings account that pays 6% interest per year. If you make no additional deposits or withdrawals, how much money will be in the account on January 1, 2028?

PV of a lump sum farther in the future

- (3)** Your aunt promises to give you \$2,000 five years from today to help with graduate school. If the discount rate is 8% per year, what is the present value today of that \$2,000?

FV of a single amount over a longer horizon

- (4)** On January 1, 2025, you invest \$1,500 in a mutual fund that you expect will earn 12% per year. If your expectations are correct and you leave the money invested, how much will your investment be worth on January 1, 2030?

Q5-Q8: NPV questions with a single outflow and a single inflow

Simple NPV, 1 outflow at $t=0$, 1 inflow later

- (5)** Imagine that your catering business were considering buying a used delivery van. The van costs \$400 today and is expected to generate a single resale value (or “salvage value”) of \$700 three years from today. If the discount rate is 12% per year, what is the Net Present Value (NPV) of buying the van, calculated as of today (year 0)?

NPV with a short horizon

- (6)** A local coffee shop can pay \$2,500 today to install a new espresso machine. The owner expects this machine to allow the shop to sell enough extra drinks that she can sell the machine used for \$3,500 in two years. If the discount rate is 15% per year, what is the NPV of purchasing the new machine today?

Outflow in one year, inflow later

- (7)** A streaming startup offers you a deal: you agree to pay \$1,000 one year from today in exchange for a \$1,600 payment four years from today (you pay nothing today). If the relevant discount rate is 10% per year, what is the NPV of this deal today (year 0)? (*Pro tip: First discount both cash flows back to today.*)

Outflow today, inflow farther in the future

- (8)** Suppose a small ecommerce brand can spend \$600 today on a new website design that will allow the firm to sell the site and associated customer list to a larger competitor for \$900 four years from today. If the discount rate is 8% per year, what is the NPV of making this investment today?

Q9-Q12 NPV questions with multiple cash flows in and out

(Two outflows, one later inflow – similar structure to Q1)

(9) A gym is considering launching a new mobile app for its members. Developing and launching the app would require:

- A payment of \$300 today (year 0) for initial development costs, and
- A payment of \$300 one year from today (year 1) for additional features and bug fixes.

The gym expects to sell the app and underlying code to another chain for \$1,000 four years from today (year 4). If the appropriate discount rate is 10% per year, what is the NPV of this project today (at year 0)?

Three outflows, two inflows

(10) A small manufacturer is evaluating a new piece of equipment. The project would require the following cash flows:

- pay \$500 today (year 0) to purchase the equipment
- pay \$400 one year from today (year 1) for installation and training
- pay \$300 two years from today (year 2) for a major upgrade.

The equipment is expected to generate:

- additional net cash inflows of \$800 three years from today (year 3), and
- a resale (“salvage”) value of \$900 four years from today (year 4)

If the discount rate is 12% per year, what is the NPV of this investment as of today (year 0)? (Please assume all cash flows occur at the end of each year.)

Three outflows, two inflows

(11) A small marketing agency is considering a multi-year advertising campaign for a client, where the agency must invest first, then expects higher fees later. The expected cash flows are:

- Outflows: \$200 today (year 0), \$200 one year from today (year 1), and \$200 two years from today (year 2).
- Inflows: \$400 three years from today (year 3) and \$400 four years from today (year 4) in higher net fees from the client.

If the appropriate discount rate is 8% per year, what is the NPV of this campaign as of today (year 0)?

Mix of early outflows and later inflows with a business context

(12) A local restaurant chain is considering a social media and loyalty-program initiative with the following expected cash flows:

- pay \$500 today (year 0) to design and build the loyalty app.
- pay \$300 one year from today (year 1) to promote the program and refine the app.

In return, the chain expects higher net profits (after all costs) of:

- \$400 two years from today (year 2),
- \$400 three years from today (year 3), and
- \$400 four years from today (year 4)

If the discount rate is 10% per year, what is the NPV of this initiative as of today (year 0)?

ANSWER GUIDE

Q1-Q4: Single cash flow questions

PV of a single future payoff

- (1) Suppose you expect to receive a \$500 bonus from your internship three years from today. If the appropriate discount rate is 10% per year, what is the present value today of that \$500?

Present value of \$500 received in 3 years at 10%

$$\text{Formula: } PV = FV / (1 + r)^3$$

$$PV = 500 / (1.10)^3$$

$$PV = 500 / 1.331 = \mathbf{\$375.66}$$

FV of a single investment

- (2) On January 1, 2025, you deposit \$800 into a savings account that pays 6% interest per year. If you make no additional deposits or withdrawals, how much money will be in the account on January 1, 2028?

Future value of \$800 invested for 3 years at 6%

$$\text{Formula: } FV = PV \times (1 + r)^3$$

$$FV = 800 \times (1.06)^3$$

$$FV = 800 \times 1.191016 = \mathbf{\$952.81}$$

PV of a lump sum farther in the future

- (3) Your aunt promises to give you \$2,000 five years from today to help with graduate school. If the discount rate is 8% per year, what is the present value today of that \$2,000?

Present value of \$2,000 received in 5 years at 8%

$$PV = 2000 / (1.08)^5$$

$$PV = 2000 / 1.46933 = \mathbf{\$1,361.17}$$

FV of a single amount over a longer horizon

- (4) On January 1, 2025, you invest \$1,500 in a mutual fund that you expect will earn 12% per year. If your expectations are correct and you leave the money invested, how much will your investment be worth on January 1, 2030?

Future value of \$1,500 invested for 5 years at 12%

$$FV = 1500 \times (1.12)^5$$

$$FV = 1500 \times 1.76234 = \mathbf{\$2,643.51}$$

Q5-Q8: NPV questions with a single outflow and a single inflow

Simple NPV, 1 outflow at $t=0$, 1 inflow later

- (5) Imagine that your catering business were considering buying a used delivery van. The van costs \$400 today and is expected to generate a single resale value (or “salvage value”) of \$700 three years from today. If the discount rate is 12% per year, what is the Net Present Value (NPV) of buying the van, calculated as of today (year 0)?

NPV: Pay \$400 today; receive \$700 in 3 years at 12%

$$\text{PV of inflow} = 700 / (1.12)^3 = 700 / 1.404928 = 498.25$$

$$\text{NPV} = -400 + 498.25 = \mathbf{\$98.25}$$

NPV with a short horizon

- (6) A local coffee shop can pay \$2,500 today to install a new espresso machine. The owner expects this machine to allow the shop to sell enough extra drinks that she can sell the machine used for \$3,500 in two years. If the discount rate is 15% per year, what is the NPV of purchasing the new machine today?

NPV: pay \$2,500 today; receive \$3,500 in 2 years at 15%

$$\text{PV of inflow} = 3500 / (1.15)^2 = 3500 / 1.3225 = 2646.50$$

$$\text{NPV} = -2500 + 2646.50 = \mathbf{\$146.50}$$

Outflow in one year, inflow later

- (7) A streaming startup offers you a deal: you agree to pay \$1,000 one year from today in exchange for a \$1,600 payment four years from today (you pay nothing today). If the relevant discount rate is 10% per year, what is the NPV of this deal today (year 0)? (*Pro tip: First discount both cash flows back to today.*)

NPV: pay \$1,000 in 1 year; receive \$1,600 in 4 years at 10%

$$\text{PV of outflow} = -1000 / 1.10 = -909.09$$

$$\text{PV of inflow} = 1600 / (1.10)^4 = 1600 / 1.4641 = 1092.82$$

$$\text{NPV} = -909.09 + 1092.82 = \mathbf{\$183.73}$$

Outflow today, inflow farther in the future

- (8) Suppose a small ecommerce brand can spend \$600 today on a new website design that will allow the firm to sell the site and associated customer list to a larger competitor for \$900 four years from today. If the discount rate is 8% per year, what is the NPV of making this investment today?

NPV: pay \$600 today; receive \$900 in 4 years at 8%

$$\text{PV of inflow} = 900 / (1.08)^4 = 900 / 1.36049 = 661.53$$

$$\text{NPV} = -600 + 661.53 = \mathbf{\$61.53}$$

Q9-Q12 NPV questions with multiple cash flows in and out

(Two outflows, one later inflow – similar structure to Q1)

(9) A gym is considering launching a new mobile app for its members. Developing and launching the app would require:

- A payment of \$300 today (year 0) for initial development costs, and
- A payment of \$300 one year from today (year 1) for additional features and bug fixes.

The gym expects to sell the app and underlying code to another chain for \$1,000 four years from today (year 4). If the appropriate discount rate is 10% per year, what is the NPV of this project today (at year 0)?

NPV: pay \$300 today; pay \$300 in 1 year; receive \$1,000 in 4 years at 10%

$$\text{PV}(-300 \text{ in year 1}) = -300 / 1.10 = -272.73$$

$$\text{PV}(1000 \text{ in year 4}) = 1000 / 1.4641 = 682.99$$

$$\text{NPV} = -300 - 272.73 + 682.99 = \mathbf{\$110.29}$$

Three outflows, two inflows

(10) A small manufacturer is evaluating a new piece of equipment. The project would require the following cash flows:

- pay \$500 today (year 0) to purchase the equipment
- pay \$400 one year from today (year 1) for installation and training
- pay \$300 two years from today (year 2) for a major upgrade.

The equipment is expected to generate:

- additional net cash inflows of \$800 three years from today (year 3), and
- a resale (“salvage”) value of \$900 four years from today (year 4)

If the discount rate is 12% per year, what is the NPV of this investment as of today (year 0)?
(Please assume all cash flows occur at the end of each year.)

NPV with multiple cash flows at 12%

Year 0: -500

Year 1: -400

Year 2: -300

Year 3: +800

Year 4: +900

$$\text{PV}(-400 \text{ at year 1}) = -400 / 1.12 = -357.14$$

$$\text{PV}(-300 \text{ at year 2}) = -300 / 1.2544 = -239.15$$

$$\text{PV}(800 \text{ at year 3}) = 800 / 1.40493 = 569.33$$

$$\text{PV}(900 \text{ at year 4}) = 900 / 1.57352 = 572.07$$

$$\text{NPV} = -500 - 357.14 - 239.15 + 569.33 + 572.07 = \mathbf{\$45.09}$$

Three outflows, two inflows

(11) A small marketing agency is considering a multi-year advertising campaign for a client, where the agency must invest first, then expects higher fees later. The expected cash flows are:

- Outflows: \$200 today (year 0), \$200 one year from today (year 1), and \$200 two years from today (year 2).
- Inflows: \$400 three years from today (year 3) and \$400 four years from today (year 4) in higher net fees from the client.

If the appropriate discount rate is 8% per year, what is the NPV of this campaign as of today (year 0)?

NPV with multiple cash flows at 8%

Year 0: -200

Year 1: -200

Year 2: -200

Year 3: +400

Year 4: +400

$PV(-200 \text{ at year 1}) = -200 / 1.08 = -185.19$

$PV(-200 \text{ at year 2}) = -200 / 1.1664 = -171.50$

$PV(400 \text{ at year 3}) = 400 / 1.25971 = 317.60$

$PV(400 \text{ at year 4}) = 400 / 1.36049 = 294.00$

$NPV = -200 - 185.19 - 171.50 + 317.60 + 294.00 = \text{\$54.89}$

Mix of early outflows and later inflows with a business context

(12) A local restaurant chain is considering a social media and loyalty-program initiative with the following expected cash flows:

- pay \$500 today (year 0) to design and build the loyalty app.
- pay \$300 one year from today (year 1) to promote the program and refine the app.

In return, the chain expects higher net profits (after all costs) of:

- \$400 two years from today (year 2),
- \$400 three years from today (year 3), and
- \$400 four years from today (year 4)

If the discount rate is 10% per year, what is the NPV of this initiative as of today (year 0)?

NPV with multiple cash flows at 10%

Year 0: -500

Year 1: -300

Year 2: +400

Year 3: +400

Year 4: +400

$PV(-300 \text{ at year 1}) = -300 / 1.10 = -272.73$

$PV(400 \text{ at year 2}) = 400 / 1.21 = 330.58$

$PV(400 \text{ at year 3}) = 400 / 1.331 = 300.83$

$PV(400 \text{ at year 4}) = 400 / 1.4641 = 273.21$

$NPV = -500 - 272.73 + 330.58 + 300.83 + 273.21 = \text{\$131.58}$